

Comparative Deep Learning and Machine Learning Analysis for Predicting Accurate Reconstruction of Virulence Genes in *Helicobacter pylori* Genome Assemblies

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Abstract

A primary gastric pathogen that causes peptic ulcers, chronic gastritis: *Helicobacter pylori*. It is crucially dependent on the correct genomic reconstruction of important virulence factors based on the sequences. Most genome assembly pipelines, however, tend to make assembly errors that lead to erroneous, discontinuous and/or shortened gene sequences, which negatively influences downstream biological interpretation. In this research, a systematic comparative analysis of four machine learning and deep learning models, such as Random Forest, XGBoost, one-dimensional convolutional neural network (CNN), and bidirectional long short-term memory (BiLSTM), was performed for the automatic classification of the quality of virulence gene reconstruction. A representative collection of 729 non-redundant sequences from 346 strains of *H. pylori* from NCBI RefSeq and GenBank was assembled, including sequences of the genes *cagA*, *vacA*, *babA* and the *hop* gene family. A rule system based on principles of molecular biology were used to assign each sequence to either complete, fragmented, or incomplete. The feature representation comprised 84 handcrafted biological descriptors (structural features and tri-nucleotide frequencies) and 768-dimensional contextual embeddings from DNABERT-2, which resulted in a final feature matrix of 852 dimensions. The imbalanced data was tackled by means of Synthetic Minority Oversampling Technique (SMOTE); models were tested using stratified 5-fold CV. The model with best individual model performance was XGBoost with an Accuracy of 0.9945, a Macro F1 of 0.8622 and a ROC-AUC of 0.9838. The best overall performance (Accuracy: 0.989, Balanced Accuracy: 0.957, Macro F1: 0.899) was achieved by the stacking ensemble of XGBoost, BiLSTM with a LR meta-learner. The biological interpretability of the framework was validated by the fact that internal stop codons, the feature that discriminated most, was consistently found by SHAP-based explainability analysis.

Keywords: *Helicobacter pylori*, genome assembly quality, DNABERT-2, XGBoost, BiLSTM, SHAP, stacking ensemble.

I. INTRODUCTION

Helicobacter pylori is a gram-negative microaerophilic bacterium which resides in the gastric mucosa of humans and has been placed in the Group I category of the World Health Organisation for carcinogenic organisms. Endemic to about 50% of the human population, and the most common aetiological cause of chronic gastritis, peptic ulcer disease and gastric adenocarcinoma. A group of well characterised virulence factors play a major role in the pathogenicity of *H. pylori*. These include the cytotoxin-associated gene A (*cagA*), the vacuolating cytotoxin A (*vacA*), the blood group antigen binding adhesin (*babA*) and the outer membrane protein encoding *hop* gene family.

Next generation sequencing (NGS) technologies have allowed for large-scale genomic studies of *H. pylori* in a wide variety of clinical and geographic contexts. Once researchers sequence a bacterial strain, computer programs (pipelines) combine millions of short pieces of DNA into long pieces and then into a map of the entire genome. Although significant progress has been made, these assembly techniques are not entirely ideal, especially for organisms such as *H. pylori* which have large genomes, high recombination rates and lots of repetitive elements.

This means that virulence genes assembled in a genome may be of three types of quality, reconstructed. A **complete** gene will have the correct start codon, a correct stop codon, and the appropriate reading frame of the correct length. A **fragmented** gene has ambiguous nucleotide sequences (N bases) or stop codons within the gene, which means it has been misassembled, with two separate sequences being accidentally connected. An **incomplete** gene does not contain a start or stop codon, or the length is considerably shorter than the standard gene of the reference to indicate that the locus was not fully sequenced.

The impact of such errors on biology is not insignificant. In some assemblies, the *cagA* gene may be fragmented, which could be the reason why the strain is considered CagA-negative and its virulence potential underestimated. Manual screening of each gene in hundreds of assembled strains is not scalable. Thus, automated quality prediction tools are key.

Rule based conserved ortholog benchmarks (e.g. BUSCO) do exist in existing bioinformatic tools, but these benchmarks do not provide a probabilistic output, have limited generalisation for novel sequences and lack the interpretability needed to provide mechanistic biological insight. Machine Learning (ML) frameworks provide an attractive alternative, especially when augmented with robust feature representations through pre-trained Genomic Language Models (GLM).

This work aims to fill these gaps by proposing an adaptation of the ML and DL pipeline by combining handcrafted biological features with embeddings produced by the transformer model DNABERT-2 for the classification of the reconstruction quality for virulence genes of *H. pylori*. The framework also uses SHAP based explainability to inform on the most important genomic features impacting on model decisions, and a stacking ensemble to ensure maximum predictive robustness.

II. LITERATURE SURVEY

In recent years, computational genomics has improved and made it possible to use machine learning and deep learning methods in biological sequence analysis, genome assembly validation, and the classification of genes. Data-driven approaches that can identify complex genomic patterns are desired, as traditional rule-based approaches are not easily scalable and generalizable.

Recently, **Ji et al. (2021)** created a transformer-based language model trained on genomic sequences with k-mer tokenisation strategies, named as DNABERT. In addition, it showed that bidirectional transformer architectures can learning the contextual biological information encoded in DNA sequences and perform well in a variety of prediction tasks, such as promoter and splice-site prediction. The work was mainly on binary genomic classification tasks, however, there was no attention given to multi-class assembly quality prediction problems, handling class imbalance and ensemble learning strategies.

XGBoost was proposed by **Chen and Guestrin (2016)** and is an efficient gradient-boosted decision tree framework which proved to be widely used and capable of good prediction for structured biological and genomic datasets because of its scalability. XGBoost works well for tabular data, but its performance relies significantly on representations of the data, and it is not contextually aware of the relationships in biological sequences.

Zeng et al. (2022) investigated the use of deep learning techniques that integrate Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks in predicting genomic features. Their CNN-LSTM model was able to capture both local and sequential biological patterns, and performed better in sequence classification tasks with higher macro F1-scores. However, these pre-trained genomic language models, explainability analysis, and imbalance correction methods were not included in the study.

Shen et al. (2018) used Bidirectional Long Short-Term Memory (BiLSTM) networks for biological sequence classification based on one-hot encoded DNA sequences. The study revealed that recurrent neural networks can capture the dependencies in series data of biological systems. One-hot encoding was however poor in terms of the representation of the context, and the prediction accuracy of minority classes was also low because of class imbalance.

BUSCO is a widely-used rule-based tool for determining the completeness of genome assembly by assessing the presence of conserved single-copy orthologs developed by Simão et al. (2015). BUSCO is highly relevant to the quality evaluation of assemblies as it classifies assemblies according to whether they are complete, fragmented, or incomplete. But the whole structure is based on the pre-existing biological rules and doesn't have machine learning, prediction at a probability level, nor does it include mechanisms for explainability.

In the present study, the advantages of these methods are put together: Handcrafted biological descriptors are combined with the contextual embedding from the pre-trained model DNABERT-2 and multiple machine learning and deep learning architectures are tested for the quality prediction of virulence gene reconstruction in the bacterium *Helicobacter pylori*. Moreover, to enhance the robustness, interpretability, and performance on minority-class prediction, imbalance correction using SMOTE and SHAP explainability analysis are integrated.

III. MATERIALS AND METHODS

A. Dataset Description

The dataset was created from the genomes of *Helicobacter pylori* in two publicly available resources, NCBI RefSeq and GenBank, which provide high quality curated and more strain diversity, respectively. Thirty-four six strain types of *H. pylori* were downloaded. The annotated genome flat file (GBFF format) for each strain was parsed to obtain the coding sequences (CDS) that matched the target virulence genes: *cagA*, *vacA*, *babA* and all genes with a prefix of 'hop' in the gene annotation (hop genes). DNA sequences were merged into a single multi-sequence FASTA file. A set of 3,716 sequences was initially collected. Owing to the highly conserved nature of virulence loci, many of these sequences were almost identical from one strain to the next (95–99% identity). Keeping all these sequences would lead to data duplication in models, over-optimistic performance figures and overfitting instead of real generalisation. Redundancy reduction was thus carried out with CD-HIT at an 95% identity level, removing sequences with 95% or more homology against each other. A final curated data set resulted in 729 non-redundant sequences (as shown in Table I, a reduction of 80.3% from the raw data pool.

TABLE I: Dataset Statistics After Redundancy Removal

Metric	Value
Initial sequences	3,716
Sequences removed (redundant)	2,987
Final unique sequences	729
Reduction (%)	80.3%

B. Auto-Labeling Pipeline

There is no existing annotated dataset of the quality of virulence gene assembly for *H. pylori* so an automated labeling system is generated based on rule and principles of molecular biology. Three quality labels were automatically assigned to each sequence, based on a coding program and without manual annotation. The labels are prioritized, starting from the most biologically relevant, as in Table II:

TABLE II: Label Distribution

Label	Number of Sequences
Complete sequences	672 (92.2%)
Fragmented sequences	49 (6.7%)
Incomplete sequences	8 (1.1%)

Priority 1 — Fragmented: The first priority of checking is to see if the sequence contains any internal stop codons and a missing start codon, which should be regarded as fragmented. A sequence is considered as fragmented if it has one or more in-frame internal stop codons (TAA, TAG, TGA in the internal frame) or it has one or more ambiguous N bases, which means it is an unresolved assembly gap.

Priority 2 — Incomplete: If the sequence is passed the fragmentation check, it will be evaluated. Any sequence that does not contain a canonical start codon (ATG, TTG, or GTG) at position 0, or a stop codon (TGA, TAA, or TAG) at the end, or that has a length ratio less than 0.80 when compared to the length of the reference gene, is considered incomplete.

Priority 3 — Complete: A complete sequence should contain a correct start codon, have a proper stop codon, not contain an internal stop codon, not contain N bases and have a length ratio of at least 0.80.

The definitions of the three labels are summarised in Table III.

TABLE III: Sequence Quality Labels and Their Definitions

Label	Definition
Complete	Full-length gene with correct start and stop codon, no internal interruptions, length ratio ≥ 0.80
Incomplete	Missing start/stop codon OR sequence length ratio < 0.80 relative to canonical length
Fragmented	Contains in-frame internal stop codons OR ambiguous N bases indicating clear assembly error

C. Feature Engineering

To make the DNA sequences interpretable by machine learning models, a 84 dimensional numerical feature vector was computed for each sequence.

Features were organised into two groups:

Group A: Structural Features (20 features): These features reflect the overall integrity of the genes and their structural validity. These are presence of start codon (binary), presence of stop codon (binary), the number of in-frame stop codons, the number of N bases, the length ratio (actual length - canonical length), GC content (proportion of G and C bases; expected 39% for *H. pylori*), raw sequence length (base pairs), and a one-hot encoding of gene identity across 13 genes of interest.

Group B: Tri-nucleotide Frequency Features (64 features). The ribosome reads the DNA in triplets (codons), so Group B: Tri-nucleotide Frequency Features (64 features). The 4 bases (A, T, G, C) can form $4^3 = 64$ different combinations of three bases (trinucleotide). The relative frequency of each trinucleotide was calculated for each sequence as the number of trinucleotide occurrences divided by the total number of trinucleotides in the sequence with the exception of ambiguous N bases. A complete gene has a consistent usage of trinucleotides which is characteristic, while a fragment (potentially an incorrect join of two unrelated sequences) has mixed or noisy trinucleotide usage.

D. DNABERT-2 Embeddings

The DNABERT-2 pre-trained transformer model was used to support the context-specific sequence representation of the 84 handcrafted features. The DNABERT-2 is a bidirectional encoder trained on massive genomic sequence collections with a multi-species approach (as BERT is in NLP). It acts on sequences of DNA as tokenised inputs and yields contextual representation of each position.

In this study, each sequence was tokenised and fed into the DNABERT-2 model. The embedding output of the [CLS] token was used as the sequence-level representation, which is a 768-dimensional vector that represents the full sequence of tokens. This resulted in an embedding matrix of size 729×768 . The 768-dimensional DNABERT embeddings were then combined with 84 handcrafted biological features to create a final combined feature matrix of shape 729×852 .

E. Workflow and System Architecture

The entire process of the proposed *H. pylori* virulence gene reconstruction quality prediction framework is shown in Figure 1. Data collection starts with data from NCBI RefSeq and GenBank, genomic data for 346 *H. pylori* strains are downloaded and then redundancy is removed using CD-HIT to produce 729 non-redundant sequences. In the second step, a biological based automatic labeling pipeline classifies each sequence as complete, fragmented or incomplete due to the presence of internal stop codons, ambiguous N bases, invalid start/stop codons, and sequence length ratios. The third stage carries out feature engineering, extracting 84 handcrafted biological features such as structural descriptors and tri-nucleotide frequency patterns. These features are then mixed with 768 dimensional contextual embeddings generated with the help of DNABERT-2 to create a final feature matrix of size 852. Four models (Random Forest, XGBoost, CNN and BiLSTM) are trained using SMOTE balancing, StandardScaler normalisation and stratified 5-fold cross-validation in the model training stage. Model behaviour is then explained using SHAP and the most important biological features are identified that account for predictions. Finally, a stacking ensemble model is used together with Logistic Regression meta-learner and XGBoost with BiLSTM model to improve robustness and classification performance of the model, which provides maximum classification accuracy when compared to the others.

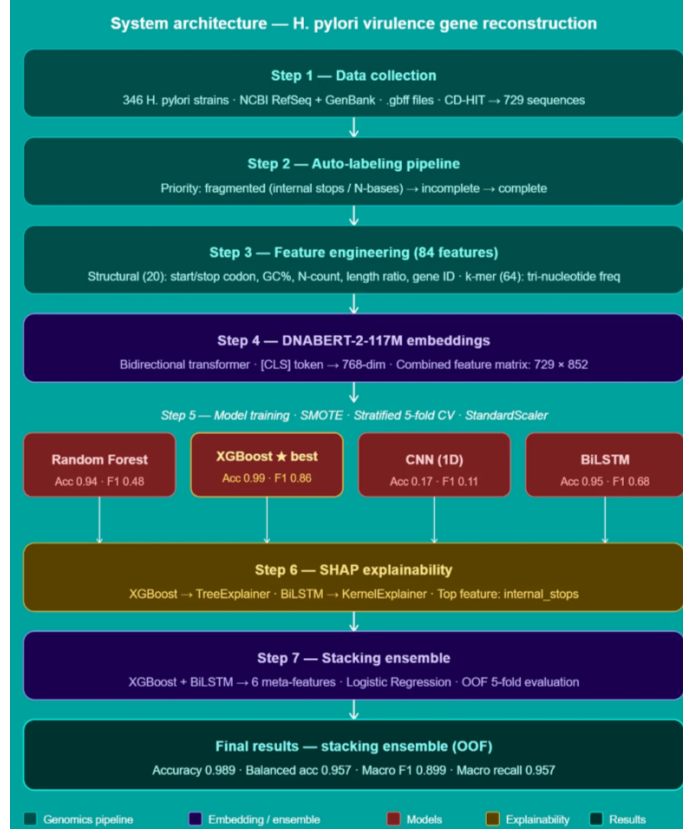


Fig. 1: System Architecture

F. Model Training

1) *Class Imbalance Handling*: There is a high class imbalance in the data—672 (92.2%) complete sequences, 49 (6.7%) fragmented, and 8 (1.1%) are incomplete. If they did not correct this distribution, classifiers learned on it would be biased towards the majority class. Two complementary approaches were employed: (i) class weights were calculated based on the inverse of class frequency and fed to the models that used them; and (ii) Synthetic Minority Oversampling Technique (SMOTE) was performed in every training fold to create synthetic samples of the minority classes, thus equalizing the class sizes prior to model fitting.

2) *Cross-Validation*: All throughout, stratified 5-fold cross-validation was used, where the proportion of classes in the training set and the testing set remained the same in each fold. To avoid data leakage, feature scaling (zero mean, unit variance) was done within each fold.

3) *Random Forest*: Random Forest is an ensemble of bagging decision trees. A Bootstrap of training data is used to train each tree, and splits are chosen from a random subset of features, decreasing overfitting and variance. The 852 dimensional reduced feature matrix is used for training Random Forest with SMOTE balanced training sets and class weights.

4) *XGBoost*: XGBoost is a regularised gradient boosting algorithm. One by one, trees are added to the ensemble, each correcting the errors of the ensemble until that point. XGBoost was trained with SMOTE balanced training data with the same class weights, and tested following the same stratified five-fold setup.

5) *CNN (1D Convolutional Neural Network)*: The 852-dimensional feature vector was used as a one dimensional input representation in a one dimensional CNN. The local feature interactions were extracted

by convolution and pooling layers, and then classified by dense layers. To avoid overfitting, early stopping was utilized by examining validated loss.

6) *BiLSTM*: The Bidirectional Long Short-Term Memory network is used to process the feature vector in both the forward and backward sequential direction, capturing the long-range dependencies. The 852 dimensional feature vector was split into four segments of 213 dimensions to allow Bidirectional LSTM to be applied. Dropout regularisation and early stopping were used during the training.

G. SHAP Explainability

For SHAP values, the XGBoost model and BiLSTM model were both calculated. SHAP does this by measuring the individual contribution of each feature towards each prediction so that both local (per sample) and global (aggregate) interpretability can be obtained. For the tree model, XGBoost was applied to 146 test samples and 852 features and the TreeExplainer was used that is exact and computationally efficient. BiLSTM applied the KernelExplainer (model-agnostic, 100 background samples $\times$ 852 features). The results were presented in the form of beeswarm plots per model and combined bar chart for direct comparison across the models.

Figure 2 shows the SHAP explainability workflow.

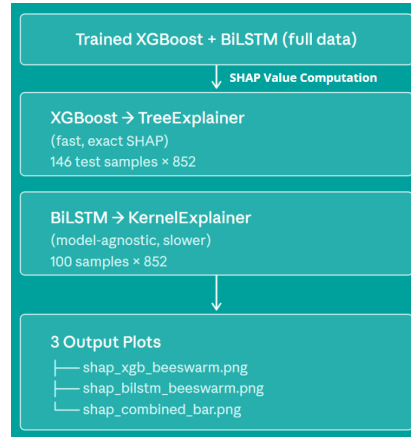


Fig. 2: SHAP Explainability

H. Stacking Ensemble

In order to get better performance, a stacking ensemble was built, which is composed of the two best performing models. The stacking pipeline is shown in Figure 3.

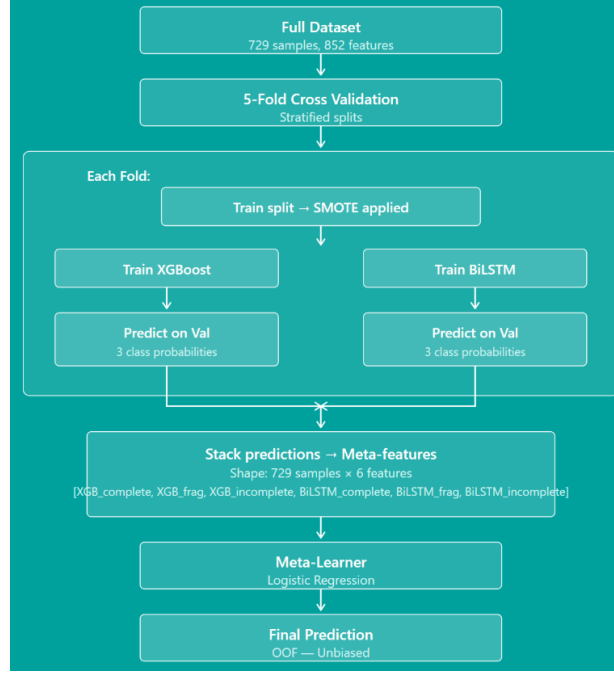


Fig. 3: Stacking Ensemble Architecture

For the five folds, XGBoost and BiLSTM were trained using the SMOTE-balanced training split, and predicted the three-class probabilities on the held-out validation fold. These probability vectors were concatenated together to obtain 6 meta-features per sample: {XGB_complete, XGB_fragmented, XGB_incomplete, BiLSTM_complete, BiLSTM_fragmented, BiLSTM_incomplete}. The complete 729×6 out-of-fold (OOF) meta-feature matrix was then created and fed into a Logistic Regression (LR) meta-learner, which was tested using an extra stratified 5-fold cross-validation. By using this OOF procedure, the meta-learner is trained in the unbiased predictions wrt base models.

IV. RESULTS AND DISCUSSION

A. Individual Model Performance

Table IV shows the performance of all four models when stratified five-fold cross validation with SMOTE balancing is used. The accuracy, balanced accuracy, macro F1-score, macro recall, mean sensitivity, mean specificity and ROC-AUC were used to evaluate all models.

TABLE IV: Individual Model Performance

Model	Accuracy	Bal. Acc.	Macro F1	Mean Sens.	Mean Spec.	ROC-AUC
Random Forest	0.9383	0.4464	0.4807	0.4464	0.7647	0.9183
XGBoost	0.9945	0.8600	0.8622	0.8600	0.9773	0.9838
CNN	0.1702	0.3583	0.1069	0.3583	0.6873	0.5490
BiLSTM	0.9451	0.6526	0.6805	0.6526	0.8001	0.8327

The result of the model performance comparison is illustrated in Figure 4.

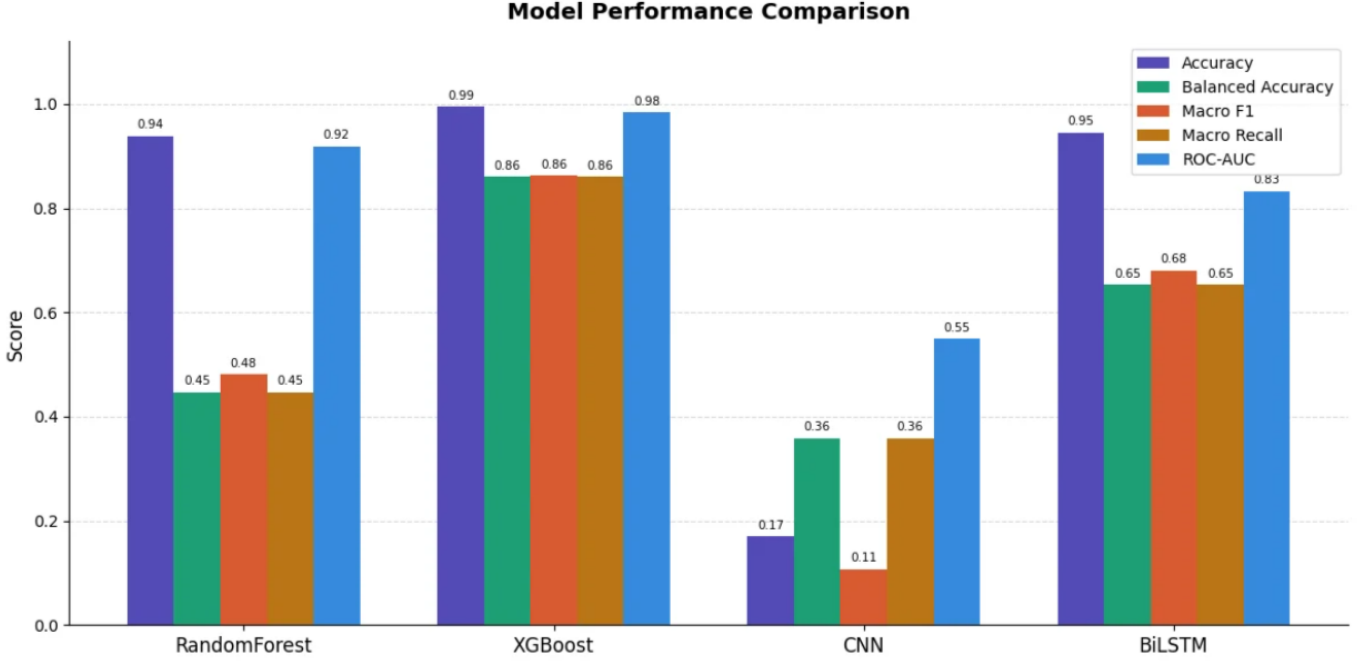


Fig. 4: Model Performance Comparison

Across all metrics, **XGBoost** had the best performance in each individual model, with an accuracy of 0.9945 and a macro F1 score of 0.8622. It was also observed that **Random Forest** showed higher overall accuracy (0.9383) but low balanced accuracy (0.4464) and macro F1 (0.4807) which indicated that it was highly skewed towards the majority class even after SMOTE augmentation. Its confusion matrix indicates 32 fragmented sequences were incorrectly classified as complete and 8 incomplete sequences were incorrectly classified, while its accuracy (0.9451) and macro F1 (0.6805) were also reasonable, indicating the ability of bidirectional recurrent networks to model the interactions of features across sequences. Performance on the incomplete class remained poor, however, due to the fact that there were very few instances in the training set, even after SMOTE, for this class and CNN is known to perform poorly when applied to non-spatial tabular feature vectors with small data sets. Under a training regime of 729 samples, the 852-dimensional vector does not contain meaningful local spatial structure that can be exploited by the convolution filters.

B. Confusion Matrices

The confusion matrices of all four models are shown in Table V.

TABLE V: Confusion Matrices of Individual Models

Model	True Class	Pred. Complete	Pred. Fragmented	Pred. Incomplete
Random Forest	Complete	667	5	0
	Fragmented	32	17	0
	Incomplete	8	0	0
XGBoost	Complete	672	0	0
	Fragmented	1	48	0
	Incomplete	3	0	5
CNN	Complete	77	595	0
	Fragmented	2	47	0
	Incomplete	1	7	0
BiLSTM	Complete	666	6	0
	Fragmented	31	18	0
	Incomplete	3	0	5

The confusion matrix of XGBoost is very clean: all 672 complete sequences are correctly classified, one fragmented sequence is mis-classed, and 5 of 8 fragment sequences are correctly classified. This is the best individual model performance recorded.

C. SHAP Explainability Analysis

The beeswarm plots of SHAP for XGBoost and BiLSTM on the fragmented class have been illustrated in Figures 5 and 6 respectively.

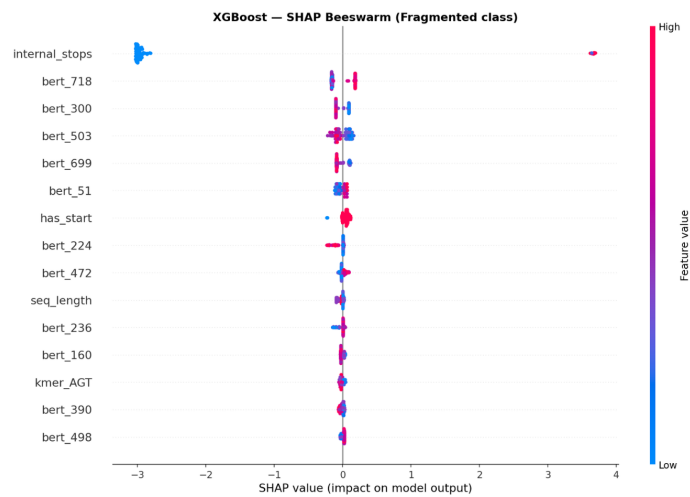


Fig. 5: XGBoost SHAP Beeswarm Plot (Fragmented Class)

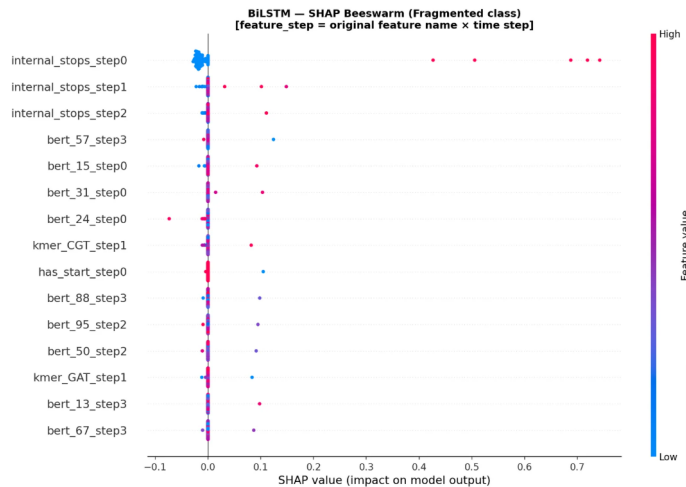


Fig. 6: SHAP Beeswarm Plot (Fragmented Class) for BiLSTM

The SHAP analysis shows that the **internal stop codons** are by far the most important biological feature for the classification of fragmented sequences, as is expected according to the labelling logic where the presence of internal stop codons is the most important indicator of assembly error. The feature is dominant in both XGBoost (TreeExplainer) and BiLSTM (KernelExplainer), which further shows the biological interpretability of both models.

Secondary contributions from the embedding dimensions of the DNABERT in the sequence level (bert_*) show that the structural information is not the only one that is important for capturing patterns, which in turn highlights the importance of combining handcrafted and learned features. Known differences in the usage of codons between complete and disrupted reading frames were supported by the presence of has_start, kmer_AGT and kmer_CGT in the top feature lists.

D. Stacking Ensemble Results

In terms of accuracy, the stacking ensemble (XGBoost + BiLSTM with Logistic Regression meta-learner) was the best with out-of-fold cross-validation. The results are displayed in Table VI and the per-class classification report is displayed in Table VII.

TABLE VI: OOF Evaluation Results for Stacking Ensemble

Metric	Value
Accuracy	0.989 ± 0.009
Balanced Accuracy	0.957 ± 0.080
Macro F1	0.899 ± 0.088
Macro Precision	0.882 ± 0.101
Macro Recall	0.957 ± 0.080

TABLE VII: Stacking Ensemble Per-Class Classification Report (Aggregated OOF)

Class	Precision	Recall	F1-Score	Support
Complete	1.00	0.99	0.99	672
Fragmented	1.00	0.98	0.99	49
Incomplete	0.54	0.88	0.67	8
Macro avg	0.85	0.95	0.88	729
Weighted avg	0.99	0.99	0.99	729

The stacking ensemble’s OOF confusion matrix is shown in Table VIII.

TABLE VIII: Confusion Matrix of Stacking Ensemble (OOF)

True Class	Pred. Complete	Pred. Fragmented	Pred. Incomplete
Complete	666	0	6
Fragmented	1	48	0
Incomplete	1	0	7

Ensemble clearly outperforms individual models in terms of balanced accuracy (0.957 vs. 0.860 for the XGBoost model alone) and macro recall (0.957 vs. 0.860 for the XGBoost model alone), suggesting that when the complementary strengths of a tree-based gradient-boosting algorithm and a bidirectional recurrent network are combined with the help of a learned meta-learner, the resulting ensemble makes for a more powerful and generalisable prediction system.

As would be predicted, the incomplete class is the one that is most difficult to classify ($F1 = 0.67$). The precision for this class is poor (0.54): 7 of the 8 are correctly identified and 6 of the 8 complete sequences are misclassified.

E. Discussion

It further highlights that XGBoost when combined with the embeddings from DNABERT-2 and SMOTE is the best standalone model to perform this task. The stacking model of XGBoost and BiLSTM achieves an even better generalisation especially in the case of minority classes. The low performance of CNN demonstrates the significance of data structure matching with the model structure. The 852 dimensional feature vector is not locally spatial enough to be suitable for convolution; the data set (729 samples) is too small for a convolutional network to learn useful filters for tabular data. The SHAP analysis has good validation of the bio foundation of the labeling system. The high level of internal stop codons is a discriminative feature that is the most similar to the Priority 1 rule of the labeling pipeline. This “explainability by data”, in which the data-learned rule is recovered from the expert-defined rule, gives confidence that the model has learned real biological signal, rather than spurious correlations. The use of DNABERT-2 embeddings greatly enhances the representation of features when compared to just using structural and k-mer features. The presence of several bert_* dimensions in the top SHAP features for multiple sequences indicates that the transformer model has learned some general properties of the sequence, including local secondary structure propensity, codon context and mutational signature, which goes beyond the explicit biological properties it was designed to learn.

V. CONCLUSION

The purpose of this research was to design and test a thorough machine learning and deep learning system to predict the quality of the reconstruction of the virulence genes for genome assemblies of *Helicobacter pylori*. 729 non-redundant sequences were curated from 346 strains, labeled using a biologically-inspired priority-based rule system, and represented by a 852 dimensional feature matrix of handcrafted structural and k-mer features, and DNABERT-2 transformer embeddings. XGBoost model had the best performance among the four models (Random Forest, XGBoost, CNN (1D), and BiLSTM) evaluated with regards to the accuracy (0.9945), Macro F1 (0.8622), and ROC-AUC (0.9838). CNN models had lower performance, because the convolutional network was not applicable to the non-spatial tabular feature structure in the limited amounts of data. All the individual models were outperformed by a stacking ensemble of XGBoost and BiLSTM with a Logistic Regression meta-learner which resulted in a better overall performance with an Accuracy of 0.989, Balanced Accuracy of 0.957, and Macro F1 of 0.899. The impacts of different biological features were validated using SHAP-based explainability analysis, which identified “internal stop codon” as the most impactful biological feature for both models, XGBoost and BiLSTM, emphasizing the biological interpretability of the framework and consistency between the impact of model learnt behavior and molecular biology principles. The study shows that quality assessment of virulence gene reconstruction is possible by automated and interpretable AI, providing an automated tool for large-scale genomic surveillance of *H. pylori* that can be scaled up for widespread use.

VI. FUTURE WORK

Any number of ways are suggested to continue this work. Per-gene analysis of the model performance would show problems in assembly and classifiability per gene, with the 13 target genes. Secondly, the existing collection of 729 sequences restricts the generalisation ability of the deep learning models, so more data, especially from more strains, and/or data augmentation techniques, would help. Third, the translation to other bacterial pathogens with critical virulence genes is a promising direction. Last but not least, incorporating long-read sequencing data (Oxford Nanopore or PacBio) may be useful to achieve further resolution in distinguishing fragmented/incomplete assemblies.

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